

****TECH-ARCH-5332****

****Department:**** Architecture

****Author:**** Edward Avila

****Effective Date:**** 2021-05-06

Architecture Reference Document: TECH-ARCH-5332

1. Document Scope and Executive Summary

This document outlines the reference architecture for the Architecture department, providing a comprehensive framework for design and development.

- * Defining the overall architecture vision and strategy
- * Outlining the core principles and guidelines for architecture design
- * Describing the standard operating procedures (SOPs) for architecture-related activities
- * Establishing compliance monitoring, penalties, and operational governance

The Architecture department is responsible for ensuring that all software applications meet the organization's quality, security, and performance standards.

2. Core Policies and Standard Operating Procedures

2.1. Architecture Design Principles

- * Separation of Concerns (SoC): modularize application logic into distinct layers
- * Single Responsibility Principle (SRP): assign specific responsibilities to each module
- * Don't Repeat Yourself (DRY): minimize code duplication
- * Keep it Simple, Stupid (KISS): prioritize simplicity and ease of maintenance

2.2. Architecture Design Guidelines

- * Use a Microservices-based architecture for scalable and maintainable applications
- * Adopt a Service-Oriented Architecture (SOA) for complex, distributed systems
- * Implement a Domain-Driven Design (DDD) approach for business domain modeling
- * Utilize a Model-View-Controller (MVC) or Model-View-ViewModel (MVVM) pattern for user interface development

2.3. Standard Operating Procedures (SOPs)

* ****Architecture Review Process****:

- + Conduct regular architecture reviews to ensure alignment with enterprise goals and standards
- + Involve stakeholders from various departments in the review process
- + Document review findings, recommendations, and action items

* ****Code Quality and Security Guidelines****:

- + Enforce coding standards for security, performance, and maintainability
- + Implement automated code analysis tools to detect vulnerabilities and errors
- + Conduct regular penetration testing and vulnerability assessments

3. Compliance Monitoring, Penalties, and Operational Governance

3.1. Compliance Monitoring

- * Regularly monitor architecture designs for compliance with enterprise-wide policies and standards

- * Conduct audits and assessments to ensure adherence to guidelines and best practices
- * Identify and address non-compliance issues promptly

3.2. Penalties for Non-Compliance

- * First-time offenders: warning and mandatory training on architecture design principles and SOPs
- * Repeat offenders: temporary suspension of architecture-related activities, followed by a comprehensive review and retraining

3.3. Operational Governance

- * Establish an Architecture Governance Board to oversee departmental operations and ensure alignment with enterprise goals
- * Define clear roles and responsibilities for architects, developers, and other stakeholders
- * Implement a change management process to manage architectural changes and ensure minimal disruption to business operations